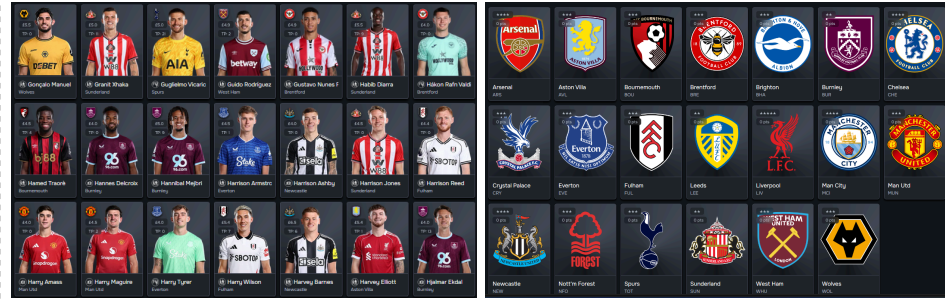


Candidate player–gameweek data



- Player, club, position, and gameweek
- Price
- Historical FPL points, minutes, and starts
- xG, xA, xGI, xGC, ICT, and selection percentage
- Availability/missing-gameweek information

Fixed-squad policy evaluation and reported outputs: GWs 27–38

- Construct a new 15-player squad once before GW27 using GWs 1–26
- Compare static and fixed-squad sequential lineup policies
- Sequential policy changes only the starting XI and bench order
- Squad membership, captain, and vice-captain remain fixed
- Formation-valid automatic substitutions
- Zero external transfers and transfer-hit deductions; no chips are used
- Calculate weekly and cumulative realized points
- Regularization, budget, Pareto, and availability sensitivity

Data preprocessing and leakage control

- Validate schema, coerce data types, deduplicate, and sort by player and GW
- Training period: GWs 1–26; evaluation period: GWs 27–38
- Treat missing player–GW observations as unavailable
- Count missing observations as zero starts for the availability estimator
- Use only lagged or pre-deadline information
- Exclude target-GW points, minutes, bonus, and bonus point system (BPS) fields
- Use price only in optimization constraints, not as a predictor

Training-period diagnostics and feature specifications

- Position-specific correlations and feature distributions
- Redundancy audit: $xGI \approx xG + xA$
- Full, Reduced, and Parsimonious Ridge specifications
- Full LASSO for automatic feature selection
- XGBoost/SHAP are used only for interpretation and not supplied to the MILP

Forecasting and cost-vector construction

1. Simple and recency-weighted averages; Holt smoothing
2. Linear trend and ARIMA variants
3. Nonparametric bootstrap and parametric Monte Carlo simulation
4. Position-specific Ridge/LASSO and hybrid recent-points–Ridge models
5. Alternative cost vectors: ICT, robust ICT, and $xGI - xGC$
6. Box-robust and availability-adjusted coefficients

Roster optimization

- **Two-stage baselines**
 - Heuristic fixed allocation: £83.5m XI and £16.5m bench
 - Actual-remainder rule: unused XI allowance passes to the bench
- **Joint 15-player MILP**
 - Select squad, starting XI, and captain simultaneously
 - Total budget £100m
- **Constraints**
 - Squad quotas: 2–5–5–3
 - Legal starting formation
 - At most three players per club
 - Captain belongs to starting XI
- **Epsilon-constraint Pareto analysis**
 - Trade-off between predicted XI-plus-captain value F_{XI} and bench value F_B

